Head-to-Tail Interference in Long-Tailed Small-Batch Training

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Abstract

Long-tailed small-batch training can update a model on frequent head classes while rare tail classes are absent. This report studies a local stability question: after two update directions make the same first-order progress on the head batch, which one causes less squared logit drift on held-out tail examples? We introduce a matched-head-gain drift diagnostic and a head-to-tail interference coefficient. For a matrix block with tail perturbation $J_T(D) = B_T D A_T$, the analysis predicts that spectral/polar geometry has a smaller worst-case drift bound when $\operatorname{nrank}(G_H) > \operatorname{ssrank}(B_T, A_T)$. Controlled boundary, long-tailed digits, and CIFAR-100-LT ResNet18 diagnostics support this local drift mechanism, while also showing that lower logit drift does not automatically imply better tail loss or accuracy.

1. Problem

Long-tailed small-batch training has a local function-drift failure mode: frequent head-class updates can change logits on rare tail examples before tail samples reappear. Ordinary loss-decrease comparisons do not isolate this effect, because an update may look stable simply by moving less. We therefore use the comparison

$$\boxed{\text{match head progress, then compare tail drift.}}$$

This is a mechanism claim rather than an optimizer-level benchmark claim: after different update geometries obtain the same first-order head-batch gain, compare how much they perturb a held-out tail function. Tail-example logit drift always means drift of the full class-logit vector on tail examples, not only tail-class logits.

2. Matched-Gain Geometry

Let $g_H = \nabla_\theta \mathcal{L}_H(\theta)$ be the head-batch gradient and let $F_T(\theta)$ be the model output on a held-out tail batch. Under a local linearization,

$$F_T(\theta + \Delta) - F_T(\theta) \approx J_T \Delta.$$

We compare updates only after fixing the same head gain,

$$-\langle g_H, \Delta \rangle = \rho,$$

and then measure tail drift $\|J_T \Delta\|$. This removes the confound that an update may look tail-safe only because it moved less and made less head progress.

For a parameter norm N, define the tail sensitivity and head-to-tail interference coefficient

$$K_{T,N} = \sup_{\|\Delta\|_N \le 1} \|J_T \Delta\|, \qquad \mathcal{I}_N(T \mid H) = \frac{K_{T,N}^2}{\|g_H\|_{N,*}^2}.$$

If Δ_N is the norm-N steepest direction scaled to head gain ρ, then

$$\|J_T \Delta_N\|^2 \le \rho^2 \mathcal{I}_N(T \mid H).$$

Thus the relevant quantity is not raw loss decrease, but tail drift after head progress has been matched. The worst-case coefficient $\mathcal{I}_N$ can be separated from the realized actual-direction quantity $\widetilde{\mathcal{I}}_N = \|J_T u_N\|^2 / \|g_H\|_{N,*}^2$, which is what direct drift measurements estimate after scaling. For non-steepest Muon-style directions, we instead use the arbitrary-direction matched-gain coefficient

$$\Delta_d = -\rho d / \langle g_H, d \rangle,$$

$$C(d; T \mid H) = \|J_T d\|^2 / \langle g_H, d \rangle^2,$$

with $\langle g_H, d \rangle > 0$. Thus $\operatorname{polar}(M_t)$ and $\operatorname{NS}(M_t)$ are normalized by their actual head alignment, not by a dual norm for the current gradient.

3. Matrix-Block Condition

For a matrix block W, write the tail perturbation as

$$J_T(D) = B_T D A_T,$$

where A_T is the tail activation entering the layer and B_T is the downstream tail Jacobian. Frobenius geometry gives

$$\mathcal{I}_F = \frac{\|B_T\|_{\mathrm{op}}^2 \|A_T\|_{\mathrm{op}}^2}{\|G_H\|_F^2},$$

while spectral geometry gives

$$\mathcal{I}_S = \frac{\sum_i \sigma_i(B_T)^2 \sigma_i(A_T)^2}{\|G_H\|_*^2}.$$

Therefore spectral geometry has a smaller worst-case matched-gain drift bound when

$$\boxed{\operatorname{nrank}(G_H) > \operatorname{ssrank}(B_T, A_T).}$$

Equivalently, $\mathcal{I}_S / \mathcal{I}_F = \operatorname{ssrank}(B_T, A_T) / \operatorname{nrank}(G_H)$. Here

$$\operatorname{ssrank}(B_T, A_T) = \frac{\sum_i \sigma_i(B_T)^2 \sigma_i(A_T)^2}{\sigma_1(B_T)^2 \sigma_1(A_T)^2}.$$

The left side measures spectral spread in the head gradient; the right side measures downstream-aware tail sensitivity, not only activation stable rank. This is a sandwich block condition. In a general nonlinear layer one should replace it with the vectorized operator form $\mathrm{vec}(\mathcal{J}_{T,\ell}(D)) = L_{T,\ell}\,\mathrm{vec}(D)$; the displayed singular-value formula is the special case $L_{T,\ell} = A_T^\top \otimes B_T$.

4. Evidence

Digits diagnostics use the scikit-learn 8×8 digits data with pixels scaled to the unit interval, a bias-free two-layer MLP with hidden dimension 32, fixed head classes 0–4, fixed tail classes 5–9, 20 random within-class splits, noisy mini-batch warmup, matched first-order gain $\rho = 0.02\mathcal{L}_H(\theta)$, and paired log-scale CIs for the tested split/seed distribution, not generalization guarantees. The MLP spectral direction is blockwise: $d_S = \{\mathrm{polar}(G_\ell)\}_{\ell=1}^2$, then globally scaled by $\Delta_d = -\rho d / \langle g_H, d\rangle$. The largest local-linearization relative-error CI upper endpoint is 2.05×10^{-3}.

The controlled boundary is directionally correct: the synthetic positive case has squared drift ratio 0.3403, while the reversed case has squared drift ratio 7.208. The controlled long-tailed digits diagnostic gives lower drift but not a direct performance claim: the one-step squared tail-example logit drift ratio is 0.5501 [0.5101, 0.5931], and the CE tail-loss difference is 0.001399 [0.0002379, 0.002559]. A GPU CIFAR-100-LT ResNet18 check gives squared drift ratio 0.5611 [0.5224, 0.6026], with a smaller-head-gain check at 0.7761 [0.7585, 0.7941], checkpoint-sweep worst ratio 0.5611 [0.5224, 0.6026], final-layer condition worst ratio 0.2391 [0.2201, 0.2597], tail-quality-control worst ratio 0.7292 [0.6923, 0.768], and all-layer ResNet JVP observed ratio 0.2011 [0.1845, 0.2192]. Head-alignment and norm ratios 2.083, 3.112, and 0.5543 show that the scaling effect is norm-specific. The Muon-style checks are compatibility tests: fixed $\mathrm{polar}(M_t)$ squared drift ratio 0.8199, trajectory $\mathrm{NS}(M_t)$ ratio 0.8019, and Fro/GD-state $\mathrm{NS}(M_t)$ ratio 0.7973. Finally, unit-JVP ratios exceed 1, but scaled layerwise ratios are 0.4766 and 0.596, so matched-head-gain scaling is part of the mechanism.

5. Interpretation

The synthetic result shows that $\mathrm{ssrank}(B_T, A_T)$ is an operative tail-side quantity: changing downstream-aware tail sensitivity changes the predicted drift ordering. The digits and CIFAR-100-LT ResNet18 one-step results give direct non-synthetic evidence for lower matched-gain tail-example logit drift, with squared drift ratios 0.5501 [0.5101, 0.5931] and 0.5611 [0.5224, 0.6026]. The Muon-style direction compatibility check reports squared drift ratios compatible with the local polar mechanism for selected momentum and finite Newton–Schulz directions, but it does not explain full Muon training dynamics.

The layerwise diagnostic rules out an overly broad interpretation. Spectral/polar unit directions are not always less tail-sensitive; in the two MLP layers, the unit-direction JVP ratios exceed 1. The lower observed drift appears after matched-head-gain scaling, meaning that spectral/polar geometry can use a smaller operator-norm step despite a larger Frobenius-norm step in this diagnostic. The mechanism is therefore not “spectral directions are always safer”, but “spectral geometry can reduce measured tail drift after head progress is matched.”

6. Claim Boundary

The supported claims are deliberately local. First, the sandwich theorem predicts a reversible boundary: the synthetic squared drift ratio changes from 0.3403 to 7.208 when the rank condition is reversed, but the alignment ablation shows that this is a bound-ordering condition rather than a standalone realized-drift predictor. Second, the digits and CIFAR-100-LT ResNet18 one-step results support a squared tail-example logit drift claim, not a broad accuracy claim. Third, Muon-style momentum directions are selected-state compatibility checks for the local polar mechanism; finite Newton–Schulz approximation, momentum-gradient alignment, and state selection remain separate sources of variation. Fourth, layerwise evidence indicates that the lower observed drift comes from matched-head-gain scaling, not from uniformly lower unit-direction tail sensitivity.

The mechanism should be evaluated before making optimizer-level claims. If spectral/polar directions lower matched-gain tail drift but do not lower tail cross-entropy or improve tail accuracy, then the result is a stability mechanism rather than a performance theorem. A practical Muon claim would require showing that the same local geometry persists under momentum, approximation, schedules, weight decay, and checkpoint variation, and that the resulting trajectory improves standard tail metrics under tuned baselines.

7. Conclusion

The current evidence supports a local mechanism claim: spectral/polar geometry can reduce matched-head-gain squared logit drift on tail examples when $\mathrm{nrank}(G_H)$ is large relative to $\mathrm{ssrank}(B_T, A_T)$, including in the CIFAR-100-LT ResNet18 one-step, checkpoint-sweep, tail-rich, and all-layer JVP checks. It does not establish optimizer-level performance or general long-tailed accuracy; the tail-rich ResNet control raises pre-update tail accuracy to 0.3739 [0.3454, 0.4024], but remains a local matched-gain diagnostic. A ResNet rank-side proxy scatter further shows why the downstream-aware term matters: mean gradient nuclear rank alone has positive correlation 0.7594 [0.6657, 0.8807] with the log drift ratio across seed/checkpoint points. The classifier-layer condition check is closer to the theorem, and the all-layer JVP control extends the local bridge to every Conv/Linear matrix weight. Stronger claims require standard long-tailed benchmarks, practical Muon ablations, and held-out predictive condition tests.

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